CURRICULUM VITAE

Dr. Stephy Jose

ADDRESS & CONTACT

Condensed Matter Theory &
Physics of Living and Non-Equilibrium Systems (PLS)
Massachusetts Institute of Technology (MIT)
Cambridge, USA
Email: sjose@mit.edu
Website: https://josestephy.github.io/

EDUCATION

Ph.D. in Physics — OCT. 2024

Tata Institute of Fundamental Research (TIFR), Hyderabad, India
Advisor: Kabir Ramola
Thesis: Hydrodynamics and Fluctuations in Models of Active Particle Motion

M.Sc. in Physics — JUN. 2019

University of Hyderabad (UOH), Hyderabad, India, CGPA 9.6/10 - *University gold medalist*

B.Sc. in Physics — MAR. 2016

Mahatma Gandhi University (MGU), Kerala, India, CGPA 9.9/10 - *University silver medalist*

RESEARCH AREAS

Non-equilibrium Statistical Physics, Active Matter, Biophysics, Phase Transitions in Living Matter, Population Genetics, Complex Systems

Topics: Active Brownian particles, bacterial motion, collective behavior, transport phenomena, evolutionary models, disordered systems, large deviations, fluctuating hydrodynamics, macroscopic fluctuation theory, exclusion processes, first passage, and extreme value problems

RESEARCH EXPERIENCE

Schmidt Science Fellow — JAN. 2026–PRESENT

Independent postdoctoral position, Massachusetts Institute of Technology (MIT), Cambridge, USA,
Hosts: Mehran Kardar and Nikta Fakhri
Self-organization in sea-sponge cell aggregates

- *Developing theoretical models of collective cell behavior in regenerating sponge aggregates.*
- *Investigating self-organization and emergent multicellular structures in living systems.*

Humboldt Research Fellow

MAR. 2025–DEC. 2025

Independent postdoctoral position, Heinrich Heine University (HHU), Düsseldorf, Germany,
Host: Hartmut Löwen
Memory Effects in Active Matter

- *Developed theoretical models of active particles with higher-order inertial and memory effects.*
- *Derived analytical results for transport, fluctuations, and emergent dynamics of chiral active systems.*

Research Associate

OCT. 2024–FEB. 2025

Tata Institute of Fundamental Research (TIFR), Hyderabad, India,
Host: Kabir Ramola
Evolutionary Models in Population Genetics

- *Derived an exact fluctuating hydrodynamic theory for the stepping-stone model, a fundamental model of spatial population genetics.*

Graduate Researcher

AUG. 2019–OCT. 2024

Tata Institute of Fundamental Research (TIFR), Hyderabad, India,
Advisor: Kabir Ramola
Hydrodynamics and Fluctuations in Models of Active Particle Motion

- *Developed theories of active matter across microscopic and macroscopic scales.*
- *Studied the interplay of activity, interactions, fluctuations and initial conditions in active systems.*

Research Assistant

JUL. 2017–JUN. 2019

University of Hyderabad (UOH), Hyderabad, India,
Advisor: P K Suresh
Inflation and T.T. Mode Angular Power Spectrum of Cosmic Microwave Background Radiation

Visiting Researcher

AUG. 2015

Inter University Centre for Astronomy and Astrophysics (IUCAA), Maharashtra, India,
Advisor: Joydeep Bagchi
Identification of Giant Radio Galaxies

Undergraduate Researcher

JUN. 2013–MAR. 2016

Mahatma Gandhi University (MGU), Kerala, India,
Advisor: Minu Joy
Superfluidity and its Applications

PUBLICATIONS

[8] R. Bhutia, **S. Jose**, P. Perlekar, and K. Ramola. 2026,
"Diffusive noise controls early stages of genetic demixing", Physical Review Letters **137**, 098401.

[7] **S. Jose*** and H. Löwen. 2025, "Jerky chiral active particles", New Journal of Physics **27**, 115003.

[6] A. Biswas, **S. Jose**, A. Pal, and K. Ramola. 2024,
"Current fluctuations in finite-sized one-dimensional non-interacting passive and active systems",
Journal of Physics A: Mathematical and Theoretical **58**, 035001.

[5] **S. Jose***, A. Rosso, and K. Ramola. 2024,
"Effect of initial conditions on current fluctuations in non-interacting active particles",
Journal of Physics A: Mathematical and Theoretical **57**, 285002.

[4] **S. Jose**, A. Rosso, and K. Ramola. 2023,
"Generalized disorder averages and current fluctuations in run and tumble particles",
Physical Review E **108**, L052601.

[3] **S. Jose***, R. Dandekar, and K. Ramola. 2023,
"Current fluctuations in an interacting active lattice gas",
Journal of Statistical Mechanics: Theory and Experiment **2023**, 083208.

[2] **S. Jose***. 2022,
"First passage statistics of active random walks on one and two dimensional lattices",
Journal of Statistical Mechanics: Theory and Experiment **2022**, 113208.

[1] **S. Jose**, D. Mandal, M. Barma, and K. Ramola. 2022,
"Active random walks in one and two dimensions", Physical Review E **105**, 064103.

* corresponding author

REFEREEING

Physical Review Letters, Physical Review E, Physical Review Research,

Journal of Statistical Mechanics, Theory and Experiment, New Journal of Physics

PROFESSIONAL SKILLS

Computation/Software: C, Fortran, Python, Mathematica, Inkscape, ImageJ

Techniques: Molecular dynamics, Monte Carlo, Finite difference, Spectral methods

Soft Skills: Oral communication, Scientific writing & editing, Mentoring & management, Website design

AWARDS

Schmidt Science Fellowship - $USD\ 110,000$/year
One of 32 fellows selected worldwide. A competitive fellowship offered by Schmidt Sciences in partnership with Rhodes Trust, designed to support early-career scientists in pursuing interdisciplinary research worldwide. 2025

Humboldt Research Fellowship $\sim EUR\ 38,000$/year
A prestigious award offered by the Alexander von Humboldt Foundation, Germany, to support researchers from around the world to conduct research in Germany. 2025

Bhagwat Saran Agarwal Memorial Gold Medal and **Prof. M. Shakuntala Memorial Gold Medal**
Ranked first overall in M.Sc. Physics (among ~ 70 candidates), University of Hyderabad, Telangana, India. 2019

Rank 2, *B.Sc. Physics* (among 1533 candidates), Mahatma Gandhi University, Kerala, India. 2016

INSPIRE Award - $INR\ 400,000$
Innovation in Science Pursuit for Inspired Research - A prestigious scholarship offered by the Department of Science and Technology (DST), India to encourage meritorious students to pursue bachelors and masters in basic sciences. 2013

State Topper, *Higher Secondary Examination*, Kerala, India (among $\sim$ 650,000 candidates); recognized among the top 1% of students nationally.

CONFERENCES, POSTERS & TALKS

Talk: *From Active Mixtures to Multicellular Assembly*, Science Leadership Program, Schmidt Science Fellows, Duke University, North Carolina, USA (Jun. 27) 2026

Talk: *Jerky Dynamics and Population Genetics*, Physics Seminar, MIT, Cambridge, USA (Jan. 30) 2026

Talk: *Jerky Chiral Active Particles*, Soft Matter Seminar, Heinrich-Heine-University Düsseldorf, Germany (Sep. 12) 2025

Talk: *Statistical Mechanics of Active Matter Across Scales*, PLS Seminar, MIT, Cambridge, USA (Jul. 1) 2025

Talk: *Hydrodynamics and Fluctuations in Models of Active Particle Motion*, Soft Matter Seminar, Heinrich-Heine-University Düsseldorf, Germany (May 10) 2024

Talk: *Introduction to Active Matter*, Alphonsa College Alumni Symposium, Kerala, India (Dec. 14) 2023

Conference: Frontiers in Statistical Physics, Raman Research Institute (RRI), Bangalore, India (Dec. 4–8) 2023

Talk: *How Do Noise and Fluctuations Play a Role in Active Systems?*, University of Hyderabad Seminar, Hyderabad, India (Sep. 20) 2023

Conference: Statistical Physics of Complex Systems, International Centre for Theoretical Sciences (ICTS), Bangalore, India (Dec. 19–23) 2022

Poster: *Active? Why Does it Matter?*, TIFR-H In-house Symposium, Tata Institute of Fundamental Research, Hyderabad, India (Sep. 15–17) 2022

School: Bangalore School on Statistical Physics XIII, International Centre for Theoretical Sciences (ICTS), Bangalore, India (Jul. 11–22) 2022

Talk: *Active Random Walks in One and Two Dimensions*, ICTS Statistical Physics Journal Club Seminar, Bangalore, India (Dec. 9) 2021

School: Bangalore School on Statistical Physics XII, International Centre for Theoretical Sciences (ICTS), Bangalore, India (Jun. 28–Jul. 9) 2021

School: Bangalore School on Statistical Physics XI, International Centre for Theoretical Sciences (ICTS), Bangalore, India (Jun. 29–Jul. 10) 2020

School: School on Gravitation and Cosmology, Inter-University Centre for Astronomy and Astrophysics (IUCAA) and Cochin University of Science and Technology (CUSAT), Kerala, India (Aug. 24–26) 2015

Workshop: Advanced Workshop on Time Domain Astronomy and Cosmology, IUCAA Resource Centre (IRC) Kochi and St. Thomas College Kozhencherry, Kerala, India (Jul. 9–11) 2015

Workshop: Workshop on Quantum Mechanics, Srinivasa Ramanujan Institute for Basic Sciences (SRIBS), Kerala, India (Jun. 20–21) 2015

Science Camp: Vigyan Jyoti Shivir (VIJYOSHI) – Annual National Science Camp, KVPY, INSPIRE Programme, IISc Bangalore and IISER Kolkata, Bangalore, India (Nov. 8–10) 2014

Science Camp: CSIR Programme on Youth for Leadership in Science (CPYLS), National Institute for Interdisciplinary Science and Technology (NIIST), Kerala, India 2011

OTHER ACHIEVEMENTS

Second prize, Poster presentation, *TIFR-H In-house Symposium*, Tata Institute of Fundamental Research, Hyderabad, India. 2022

Second prize, *All Kerala Physics Talent Search Examination (TSE 2015–2016)*, Academy of Physics Teachers (APT), Kerala, India. 2016

First prize, *Prof. A. V. Varkey Memorial Intercollegiate Physics Quiz Competition*, St. Thomas College, Kerala, India. 2016

First prize, *Prof. Shollykutty Memorial Intercollegiate Physics Presentation Contest*, St. Berchmans College, Kerala, India. 2016

Second prize, *Prof. Mohan Kurian Memorial Physics Quiz Competition*, CMS College, Kerala, India. 2016

First prize, *All Kerala Intercollegiate Physics Seminar Presentation Competition*, Devamatha College, Kerala, India. 2015

First prize, *Prof. Brigeethamma Devasia and Prof. Kunjamma Mathew Memorial Intercollegiate Physics Quiz Competition*, Assumption College, Kerala, India. 2015

Second prize, *Prof. A. V. Varkey Memorial All Kerala Intercollegiate Physics Quiz Competition*, St. Thomas College, Kerala, India. 2015

Third prize, *All Kerala Intercollegiate Seminar Competition on Recent Trends in Physics*, Alphonsa College, Kerala, India. 2015

Third prize, *Golden Jubilee Memorial Prof. Leela Philip All Kerala Intercollegiate PowerPoint Presentation Competition in Physics*, Assumption College, Kerala, India. 2015

PROFESSIONAL DEVELOPMENT

Kaufman Teaching Certificate Program, Teaching + Learning Lab, Massachusetts Institute of Technology (MIT) 2026
A semester-long series of eight practice-based workshops and microteaching sessions on evidence-based teaching, covering course design, syllabus construction, lesson planning, assessment, and inclusive classroom climate.

Interdisciplinary Science Summit, Schmidt Science Fellows, North Carolina, USA (Jun. 29–Jul. 1). *Topics: interdisciplinary science, leading university research, and AI agents* 2026

Science Leadership Program, Schmidt Science Fellows, North Carolina, USA (Jun. 25–29). *Topics: faculty careers, research collaborations, artificial intelligence, and time management.* 2026

Science Leadership Program, Schmidt Science Fellows, Northern California, USA (Feb. 8–13). *Topics: entrepreneurship, research ethics, mentoring, science communication, and interdisciplinary research.* 2026

Science Leadership Program, Schmidt Science Fellows, Oxford, UK (Oct. 5–10). *Topics: science communication, team leadership, science policy, networking, and research funding.* 2025

Annual Humboldt Meeting, Alexander von Humboldt Foundation, Berlin, Germany (Jun. 25–27). *Focused on networking with researchers, interactions with the President of Germany, and visits to leading research institutes.* 2025

EXTRACURRICULAR ACTIVITIES

Public Talk: *Disha: Rethinking Education for Millions of Minds*, Science Leadership Program, Schmidt Science Fellows, Northern California, USA. 2026

Science Outreach: *Sawaal-Jawaab*, a TIFR initiative promoting basic science through public talks and outreach activities. https://www.tifrh.res.in/events/sawaal-jawaab/ 2021–2024

Science Communication: Author of the article: *From PhD to Postdoc: My Path to the Humboldt Research Fellowship*, published in IndiaBioscience. https://indiabioscience.org/columns/stories-from-scientists/from-phd-to-postdoc-my-path-to-the-humboldt-fellowship 2025

Science Communication: Featured in the *In Focus* video series presenting graduate research at TIFR Hyderabad. https://www.youtube.com/watch?v=RfinunGbv2Q&ab_channel=TIFRHyderabad 2023

Community Service: Volunteer, *Where Are India's Children (WAIC)*, supporting initiatives to promote adoption and family-based care for vulnerable children. https://waic.in/ 2019–2026

Community Service: Volunteer, *Aikyatha*, supporting fundraising and social outreach initiatives. https://aikyatha-org.my.canva.site/ 2021–2026

Classical Dance: Professional Kuchipudi dancer; completed certificate course with first-class distinction, Sri Thyagaraja Government College for Music and Dance, Telangana, India. 2019–2024

Public Talk: *Balancing Science and Art: My Journey*, Nritya Priya Dance Academy, Telangana, India. 2024

POSITIONS OF RESPONSIBILITY

Session Chair, Conference: *Statistical Physics: Recent advances and Future directions*, The International Centre for Theoretical Sciences, Bangalore, India 2022

Grader, Course on Condensed Matter,
Tata Institute of Fundamental Research, Hyderabad, India 2021

Student Coordinator, Placement Cell,
University of Hyderabad, India 2017–2019

Student Chairperson, Physics Department,
Alphonsa College, Pala, India 2015–2016